

1

Urine Collection

N = 940 patients
9 cohorts
5 cancer types + 4 controls

2

SERS Measurement

Gold nanoparticle substrate
Portable Raman (785 nm)
5 replicates / sample

3

Two-Stage AI Classifier

Stage 1
Cancer vs Normal

Stage 2
Cancer Subtype

Cancer-Specific Spectral Signatures

